

Nombre:

Día

Mes

Año

Folio

Tema:

$$10 + 1 = 11$$

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Algebra Lineal

Grupo 17

Semestre 2022-2

Tercer examen parcial

1)

$$F((a,b), (c,d)) = ac + ad + bc + bd$$

Solución

1- Simetría

$$(\bar{x} | \bar{y}) = (\bar{y} | \bar{x})$$

$$(a,b | c,d) = (c,d | a,b)$$

$$ac + ad + bc + bd = ca + cb + da + db$$

$$ac + ad + bc + bd = ac + bc + ad + bd$$

Si:
completa

2- Distributividad

$$(\bar{x} | \bar{y} + \bar{z}) = (\bar{x} | \bar{y}) + (\bar{x} | \bar{z})$$

$$\bar{x} = (a,b) \quad \bar{y} = (c,d) \quad \bar{z} = (e,f)$$

$$[(a,b) | (c+e, d+f)] = [(a,b) | (c,d)] + [(a,b) | (e,f)]$$

$$a(c+e) + a(d+f) + b(c+e) + b(d+f) = ac + ad + bc + bd + ae + af + be + bf$$

$$ac + ae + ad + af + bc + be + bf + bf = ac + ad + bc + bd + ae + af + be + bf$$

/s:

comple

3.- Homogeneidad

$$(\alpha \bar{x} | \bar{y}) = \alpha (\bar{x} | \bar{y})$$

$$(\alpha(a, b) | (c, d)) = \alpha((a, b) | (c, d))$$

$$(\alpha a, \alpha b) | (c, d) = \alpha(ac + ad + bc + bd)$$

$$\alpha ac + \alpha ad + \alpha bc + \alpha bd = \alpha ac + \alpha ad + \alpha bc + \alpha bd$$

/s:

comple

4.- Positividad

$$(\bar{x} | \bar{x}) > 0 \Leftrightarrow \bar{x} \neq \emptyset$$

$$[(a, b) | (a, b)] = aa + ab + ba + bb$$

$$= a^2 + 2ab + b^2$$

$$= (a+b)^2$$

$$\text{Si } b = -a$$

$$(a-a)^2 = 0$$

/s:

comple

• No es un producto interno, ya que no se cumple la positividad

2)

$$H = \{(a, b, c); a, b, c \in \mathbb{R}\}$$

a) El complemento ortogonal de H

$$\left. \begin{array}{l} a=1 \\ b=0 \end{array} \right\} (1, 0, 1)$$

$$\left. \begin{array}{l} a=0 \\ b=1 \end{array} \right\} (0, 1, 0)$$

$$B = \{(1, 0, 1), (0, 1, 0)\}$$

$$W^\perp = \{ \bar{v} \in V \mid (\bar{v} | \bar{w}) = 0; \bar{w} \in W \}$$

$$W^\perp = \{ (a, b, c) \in \mathbb{R}^3 \mid (a, b, c) \mid \begin{array}{l} (1, 0, 1) \\ (0, 1, 0) \end{array} = 0 \}$$

$$(a, b, c) \mid (1, 0, 1) = a + c = 0 \rightarrow a = -c$$

$$(a, b, c) \mid (0, 1, 0) = b = 0$$

$$\therefore W^\perp = \{ (-c, 0, c) \mid c \in \mathbb{R} \}$$

Comprobación

$$(\bar{v} | \bar{w}) = 0$$

$$(-c, 0, c) \mid (a, b, a) = -ca, 0, ca$$

$$\phi = \phi$$

Si
da la comp. //

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b) La expresión del vector $\vec{D} = (1, 1, -1)$ como $\vec{U} = \vec{H} + \vec{H}^\perp$

$$(1, 1, -1) = (a, b, a) + (c, 0, c)$$

$$(1, 1, -1) = (a-c, b, a+c)$$

$$a-c=1$$

$$b=1$$

$$a+c=-1$$

Sistema
de
ecuaciones

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & -1 \end{array} \right) \begin{array}{l} R_1 - R_3 = R_1 \\ \\ \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & -2 \end{array} \right) \begin{array}{l} \\ \\ \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & -2 \end{array} \right) \begin{array}{l} \\ \\ \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \end{array} \right) \begin{array}{l} R_3 = R_3 \\ \\ \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \end{array} \right) \begin{array}{l} \\ \\ \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \end{array} \right) \begin{array}{l} \\ \\ (-1)R_3 = R_3 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \begin{array}{l} R_1 - R_3 = R_1 \\ \\ \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \begin{array}{l} \\ \\ \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \begin{array}{l} \\ \\ (-1)R_3 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right) \begin{array}{l} \\ \\ \end{array}$$

$$a=0$$

$$b=1$$

$$c=-1$$

$$(1, 1, -1) = (0 - (-1), 1, 0 + (-1))$$

$$\therefore \vec{U} = \vec{H} + \vec{H}^\perp$$

$$(1, 1, -1) = (0, 1, 0) + (1, 0, -1)$$

Comprobación

$$(1, 1, -1) = (0, 1, 0) + (1, 0, -1)$$

$$(1, 1, -1) = (1, 1, -1)$$

Si:
da b
comp

3)

$$N = \{(a, b, c); -2a + b - c = 0; a, b, c \in \mathbb{R}\}$$

$$\vec{v} = (0, 0, 1)$$

Solución

$$N = \{(a, b, -2a + b) | a, b \in \mathbb{R}\}$$

$$\left. \begin{matrix} a=1 \\ b=0 \end{matrix} \right\} (1, 0, -2)$$

$$\left. \begin{matrix} a=0 \\ b=1 \end{matrix} \right\} (0, 1, 1)$$

$$B = \left\{ \underbrace{(1, 0, -2)}_{\vec{v}_1}, \underbrace{(0, 1, 1)}_{\vec{v}_2} \right\}$$

$$\vec{w}_1 = \vec{v}_1$$

$$\vec{w}_1 = (1, 0, -2)$$

Nombre:

CUI

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$$\vec{w}_2 = \vec{v}_2 - \frac{(\vec{v}_2 | \vec{w}_1)}{(\vec{w}_1 | \vec{w}_1)} \vec{w}_1$$

$$\vec{w}_2 = (0, 1, 1) - \frac{[(0, 1, 1) | (1, 0, -2)]}{[(1, 0, -2) | (1, 0, -2)]} (1, 0, -2)$$

$$(0, 1, 1) - (1, 0, -2) = (0)(1) + (1)(0) + (1)(-2)$$

$$= 0 + 0 - 2$$

$$= -2$$

$$(1, 0, -2) | (1, 0, -2) = (1)(1) + (0)(0) + (-2)(-2)$$

$$= 1 + 0 + 4$$

$$= 5$$

$$\vec{w}_2 = (0, 1, 1) - \left[\frac{-2}{5} \right] (1, 0, -2)$$

$$\vec{w}_2 = (0, 1, 1) - \left(-\frac{2}{5}, 0, -\frac{4}{5} \right)$$

$$\vec{w}_2 = \left(0 - \left(-\frac{2}{5} \right), 1 - 0, 1 - \left(-\frac{4}{5} \right) \right)$$

$$\vec{w}_2 = \left(\frac{2}{5}, 1, \frac{9}{5} \right)$$

$$B_{ortg} = \left\{ \underbrace{(1, 0, -2)}_{\vec{b}_1}, \underbrace{\left(\frac{2}{5}, \frac{1}{5}, \frac{1}{5}\right)}_{\vec{b}_2} \right\}$$

$$\vec{b}_1 = \frac{\vec{w}_1}{\|\vec{w}_1\|}$$

$$\vec{b}_1 = \frac{(1, 0, -2)}{\|(1, 0, -2)\|}$$

$$\begin{aligned} \left[(1, 0, -2) \mid (1, 0, -2) \right]^{1/2} &= \left[(1)(1) + (0)(0) + (-2)(-2) \right]^{1/2} \\ &= \left[1 + 0 + 4 \right]^{1/2} \\ &= \sqrt{5} \end{aligned}$$

$$\vec{b}_1 = \frac{(1, 0, -2)}{\sqrt{5}} \quad \text{ó} \quad \frac{1}{\sqrt{5}} (1, 0, -2)$$

$$\vec{b}_2 = \frac{\vec{w}_2}{\|\vec{w}_2\|}$$

$$\vec{b}_2 = \frac{\left(\frac{2}{5}, \frac{1}{5}, \frac{1}{5}\right)}{\left\|\left(\frac{2}{5}, \frac{1}{5}, \frac{1}{5}\right)\right\|}$$

$$\left[\left(\frac{2}{5}, \frac{1}{5}, \frac{1}{5}\right) \mid \left(\frac{2}{5}, \frac{1}{5}, \frac{1}{5}\right) \right]^{1/2} = \left[\left(\frac{2}{5}\right)\left(\frac{2}{5}\right) + (1)(1) + \left(\frac{1}{5}\right)\left(\frac{1}{5}\right) \right]^{1/2}$$

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$$= \left[\frac{4}{25} + 1 + \frac{1}{25} \right]^{1/2}$$

$$= \left[\frac{30}{25} \right]^{1/2}$$

$$= \sqrt{\frac{30}{25}} = \frac{\sqrt{30}}{\sqrt{25}} \rightarrow \frac{5}{\sqrt{30}}$$

$$b_2 = \frac{5 \left(\frac{2}{5}, 1, \frac{1}{5} \right)}{\sqrt{30}}$$

$$= \frac{5}{\sqrt{30}} \left(\frac{2}{5}, 1, \frac{1}{5} \right)$$

$$B_{ortn} = \left\{ \underbrace{\frac{1}{\sqrt{5}} (1, 0, -2)}_{\bar{b}_1}, \underbrace{\frac{5}{\sqrt{30}} \left(\frac{2}{5}, 1, \frac{1}{5} \right)}_{\bar{b}_2} \right\}$$

$$\bar{v}_w = \sum_{i=1}^n (\bar{v} | \bar{b}_i) \bar{b}_i$$

$$\sum_{i=1}^2 (\bar{v} | \bar{b}_i) \bar{b}_i = (\bar{v} | \bar{b}_1) \bar{b}_1 + (\bar{v} | \bar{b}_2) \bar{b}_2$$

$$(\bar{v} | \bar{b}_1) \bar{b}_1$$

$$\left[(0, 0, 1) | \frac{1}{\sqrt{5}} (1, 0, -2) \right] = (0) \left(\frac{1}{\sqrt{5}} \right) + (0)(0) + (1) \left(-\frac{2}{\sqrt{5}} \right)$$

$$= 0 + 0 - \left(-\frac{2}{\sqrt{5}} \right)$$

$$= -\frac{2}{\sqrt{5}}$$

$$(\bar{v} | b_1) b_1 = \left(-\frac{2}{2\sqrt{5}} \right) \left(\frac{1}{\sqrt{5}}, 0, -\frac{2}{\sqrt{5}} \right)$$

$$(\bar{v} | b_1) b_1 = \left(-\frac{2}{5}, 0, \frac{4}{5} \right)$$

$$(\bar{v} | b_2) b_2$$

$$\left[(0, 0, 1) \right] \left(\frac{5}{\sqrt{36}} \left(\frac{2}{3}, \frac{1}{3}, \frac{1}{3} \right) \right) = (0) \left(\frac{10}{\sqrt{36}} \right) + (0) (1) + (1) \left(\frac{5}{\sqrt{36}} \right)$$

$$= 0 + 0 + \frac{5}{\sqrt{36}}$$

$$= \frac{5}{\sqrt{36}}$$

$$(\bar{v} | b_2) b_2 = \left(\frac{5}{\sqrt{36}} \right) \left(\frac{2}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

$$(\bar{v} | b_2) b_2 = \left(\frac{1}{15}, \frac{1}{6}, \frac{1}{30} \right)$$

$$\sum_{j=1}^2 (\bar{v} | b_j) b_j = \left(-\frac{2}{5}, 0, \frac{4}{5} \right) + \left(\frac{1}{15}, \frac{1}{6}, \frac{1}{30} \right)$$

$$= \sum_{j=1}^2 (\bar{v} | b_j) b_j = \left(-\frac{1}{3}, \frac{1}{6}, \frac{5}{6} \right)$$